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Carbon- and energy-efficient ethanol electrosynthesis via interfacial cation enrichment

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Ali Shayesteh Zeraati[1,5], Feng Li[1,5], Tartela Alkayyali[1,5], Roham Dorakhan[2,5], Erfan Shirzadi[2], Fatemeh Arabyarmohammadi[1], Colin P. O'Brien[1], Christine M. Gabardo[1], Jonathan Kong[3], Adnan Ozden[1,4], Mohammad Zargartalebi[1], Yong Zhao[1], Lizhou Fan[2], Panagiotis Papangelakis[1], Dongha Kim[2], Sungjin Park[2], Rui Kai Miao[1], Jonathan P. Edwards[1], Daniel Young[1], Alexander H. Ip[2], Edward H. Sargent[2] & David Sinton[1]

The use of acidic electrolytes in CO_2 reduction avoids costly carbonate loss. However, the energy efficiency of acid-fed electrolysers has been limited by high hydrogen production and operating potentials. We find that these stem from the lack of alkali cations at the catalyst surface, limiting CO_2 and CO adsorption. In acid-fed membrane electrode assembly systems, the incorporation of these cations is challenging as there is no flowing catholyte. Here an interfacial cation matrix (ICM)–catalyst heterojunction is designed that directly attaches to the catalyst layer. The negatively charged nature of the ICM enriches the alkali cation concentration near the cathode surface, trapping generated hydroxide ions. This increases the local electric field and pH, increasing multi-carbon production. Integrating the ICM strategy with a tailored copper–silver catalyst enables selective ethanol production through a proton-spillover mechanism. We report a 45% CO_2-to-ethanol Faradaic efficiency at 200 mA cm^{-2}, carbon efficiency of 63%, full-cell ethanol energy efficiency of 15% (3-fold improvement over the best previous acidic CO_2 reduction value) and energy cost of 260 GJ per tonne ethanol, the lowest among reported ethanol-producing CO_2 electrolysers.

Electrochemical CO_2 reduction (CO_2R) can utilize CO_2 emissions and produce fuels via renewable electricity. Recent research has resulted in many advancements to catalyst design, product selectivity and electrolyser configuration[1–4]. Ethanol is of particular interest among reported CO_2R products owing to its high energy density and use as a fuel. As ethanol is a platform chemical with a global market size exceeding \$100 billion yr^{-1}, the decarbonization of ethanol production is a priority[5]. Biomass-based ethanol production has solidified it as a platform chemical and fuel, but the associated carbon intensity remains high at >1.7 kg CO_2 equivalents per kilogram of ethanol (Fig. 1a). The electroreduction of CO_2 offers a promising pathway for reducing the carbon intensity associated with ethanol production, with profitability being reliant on high-performance metrics. Most advancements in ethanol electrosynthesis using CO_2R have been achieved in alkaline and neutral electrolytes[6–12]. These electrolytes facilitate an alkaline environment in the catalyst layer[13], which provides favourable conditions for

[1]Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Ontario, Canada. [2]Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada. [3]Department of Materials Science and Engineering, University of Toronto, Toronto, Ontario, Canada. [4]Department of Mechanical and Nuclear Engineering, Khalifa University, Abu Dhabi, United Arab Emirates. [5]These authors contributed equally: Ali Shayesteh Zeraati, Feng Li, Tartela Alkayyali, Roham Dorakhan. e-mail: sinton@mie.utoronto.ca

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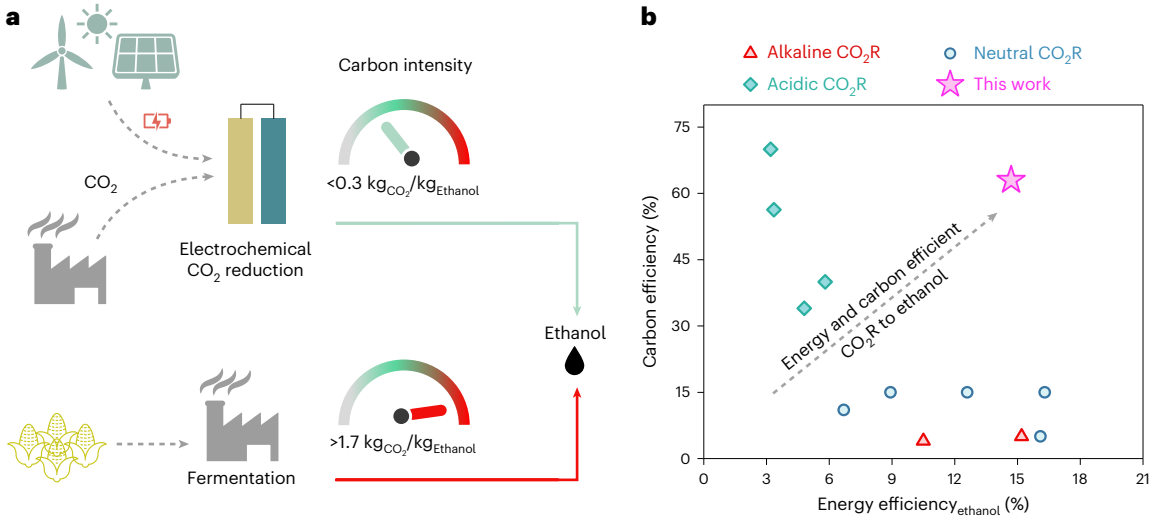

Fig. 1 | Carbon intensity and performance evaluation of CO_2R in MEA electrolysers. a, The carbon intensity for ethanol production from CO_2R and biomass[56]. The carbon intensity for the CO_2R route was calculated on the basis of the carbon intensity of renewable electricity used by the electrolyser. **b**, A comparison of carbon and energy efficiency for ethanol electrosynthesis in the benchmark alkaline, neutral and acidic CO_2 MEA electrolysers. The results are based on systems operating at current densities ≥100 mA cm^{-2}. Alkaline MEA electrolysers are reported in refs. 9,10, neutral MEA electrolysers are reported in refs. 6–8,11,12 and acidic MEA electrolysers are reported in refs. 16–18,40.

CO_2R to multi-carbon (C_{2+}) products with ethanol energy efficiencies ($EE_{ethanol}$) approaching 10% (Fig. 1b). However, the CO_2R evolves hydroxide ions (OH^-) that then combine with unreacted, dissolved CO_2 to form negatively charged (bi)carbonate ions (Supplementary Note 1 and Supplementary Fig. 1). This (bi)carbonate ion formation limits the carbon efficiency (defined as the percentage of CO_2 converted per total CO_2 input; equivalent to single-pass carbon efficiency (SPCE)) to typically below 20% towards multi-carbon products, resulting in a severe CO_2 recovery cost penalty[14,15]. Operation of CO_2R in acidic conditions can alleviate this (bi)carbonate formation issue. In acidic conditions, (bi)carbonate ion formation is mitigated by the high proton concentration in the electrolyte surrounding the cathode, enabling higher carbon efficiency (>30%)[16–18]. However, the unshielded cathode surface substantially favours the hydrogen evolution reaction (HER) over CO_2R owing to the high availability of protons and the fast kinetics of the HER, prohibiting high $EE_{ethanol}$ (Fig. 1b)[16,18]. This compromise between energy efficiency and carbon efficiency is fundamental to current alkaline, neutral and acidic approaches and motivates material and system design principles that would enable ethanol electrosynthesis with practical carbon efficiency and $EE_{ethanol}$.

To achieve a high $EE_{ethanol}$, we initially sought to increase the CO_2R Faradaic efficiency (FE) towards ethanol at a high current density and a low full-cell voltage in a carbon-efficient electrolyser. Membrane electrode assembly (MEA) electrolysers demonstrate lower full-cell voltages than their flow cell counterpart and offer a path towards industrial scalability. Operating CO_2R in an MEA with an acidic electrolyte and a cation-exchange membrane (CEM; Supplementary Fig. 1b) leads to ~100% HER and zero CO_2 crossover[16,17]. We recently demonstrated porous proton-blocking ad-layer strategies, enabling acidic CO_2R in flow cell electrolysers[19,20]. However, that strategy failed to translate to the MEA system, with the HER dominating all other reactions (Supplementary Fig. 2). The very different reaction environment present in the MEA demands a new approach[16,17,19,20].

In this study, we aimed to produce ethanol via CO_2 electrocatalysis with high carbon efficiency and $EE_{ethanol}$. In a forward-bias bipolar membrane (f-BPM) MEA operating with an acidic anolyte, we found that the lack of alkali cations (such as K^+) at the cathode surface limits CO_2 or CO adsorption on Cu, which leads to high HER and low $EE_{ethanol}$. An additional cause of reduced performance was hydroxide ion migration from the cathode surface to the anion-exchange membrane (AEM)–CEM interface of the BPM, reducing alkalinity on the cathode and, therefore, diminishing C–C coupling (a key step along the pathway to C_{2+} products). To overcome these challenges, we developed an interfacial cation matrix (ICM)–catalyst heterojunction to enrich alkali cations at the catalyst surface and enable trapping of locally generated OH^-, resulting in selectivity for CO_2R products at an FE_{CO_2R} of ≥90% and HER FE of ≤10% at 200 mA cm^{-2}. We combined the ICM strategy with a Cu–Ag catalyst designed for ethanol electroproduction, resulting in simultaneous achievement of 63% carbon efficiency, 15% full-cell $EE_{ethanol}$ and a total energy cost of 260 GJ per tonne ethanol.

Results and discussion

Local CO_2R environment under acidic conditions

To assess the catalyst surface conditions when CO_2R is operated in an acid-fed MEA, we employed finite element method (FEM) simulation (Supplementary Fig. 3). If no AEM was used, the local catalyst environment would be expected to be highly acidic due to proton accumulation at the outer Helmholtz plane (OHP), resulting in 100% HER (Supplementary Fig. 1d). Depositing an anion-exchange layer (AEL) on the catalyst surface would increase the local pH, with thicker layers favouring CO_2R (Supplementary Fig. 3). The use of an AEL maintains a local alkaline environment in acidic CO_2R, but it fails to completely shield the cathode from proton electromigration, leading to a high H_2 FE particularly at high-carbon-efficiency regimes[16,17,19]. We also found that the proton accumulation at the OHP results in a low electric field intensity (Fig. 2a), which could likewise favour HER. Therefore, we studied the effect of adding a salt to the bulk electrolyte. Alkali cations from the salt would electromigrate to the cathode and accumulate at the OHP (Fig. 2b), increasing the local electric field intensity compared with that produced when only protons are present (Fig. 2a)[21]. Previous research also showed better CO_2R selectivity as the local cathodic electric field intensity increased[22,23]. Therefore, using alkali cations in f-BPM CO_2 electrolysers could suppress the competing HER by increasing the FE towards multi-carbon products. However, high concentrations of cations need to be added to the anolyte to improve CO_2R FE, which may lead to salt precipitation and stability issues[9,24–26]. The FEM simulation demonstrated that even a low concentration of cations at the cathode surface could result in a sufficient electrical field for CO_2R

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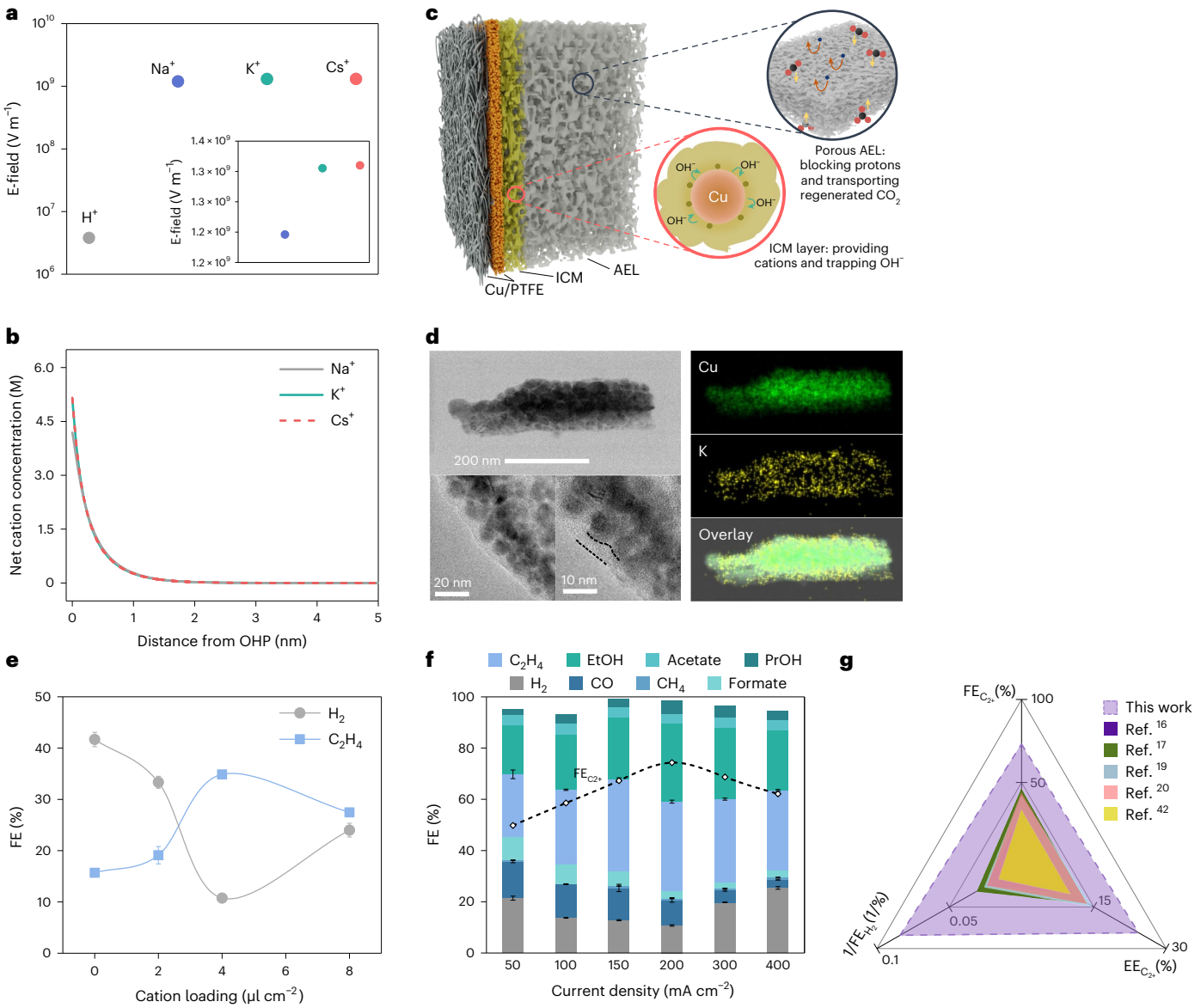

Fig. 2 | Characterization of the ICM with a Cu catalyst. a,**b**, Modelled electric field (E-field) intensity (**a**) and net cation concentration (**b**) of different cations under an applied current density of 100 mA cm^{-2}. **c**, GDE equipped with an ICM and a porous AEL. (i) The ICM provides cations close to the cathode surface and traps the generated OH$^-$; (ii) the AEL blocks the proton flux from the anode and transports the generated CO_2 to the cathode surface for further reduction. **d**, STEM with the corresponding elemental EDS mapping of K and Cu and TEM images of ICM–Cu on a PTFE nanofibre. These show a conformal coating of ICM and the presence of potassium on the Cu surface. **e**, Hydrogen and ethylene FE at different concentrations of potassium in the ICM at 200 mA cm^{-2}. The case where there is zero cation loading indicates the absence of ICM. **f**, The FE of CO_2R products and H_2 at different current densities with the corresponding C_{2+} FE (composed of ethylene, ethanol (EtOH), *n*-propanol (PrOH) and acetate) in a Cu-based MEA equipped with an ICM and 0.01 M H_2SO_4 anolyte. **g**, A comparison of FE$_{C_{2+}}$, C_{2+} energy efficiency (EE$_{C_{2+}}$) and H_2 FE of an ICM-equipped Cu GDE with the performance metrics of previously reported CO_2R electrolysers. Comparisons are constrained to reports with carbon efficiency of ≥60% (refs. 16,17,19,20,42). For **e** and **f**, the plotted values are means of three replicates, while error bars indicate the standard deviation.

(Supplementary Fig. 4). To investigate the effect of a low concentration of alkali cations on CO_2R activity, we performed surface-enhanced Raman spectroscopy measurements on Cu electrodes in contact with 0.01 M H_2SO_4 and 0.01 M H_2SO_4 + 5 mM of K_2SO_4 electrolytes in a flow cell (Supplementary Note 2 and Supplementary Figs 5 and 6). The results show that the addition of a low-concentration cation source facilitates CO_2R. By considering this observation, we sought a way to have cations present locally but not in the bulk electrolyte in an acid-fed MEA.

One approach would be to enrich the catalyst surface with cations with a multifunctional heterojunction. Perfluorinated sulfonic acid (PFSA) ionomers, such as Nafion, consist of a hydrophobic polytetrafluoroethylene (PTFE, –CF_2) backbone and a hydrophilic perfluorinated vinyl ether side-chain terminated with a sulfonate group (–SO_3^-)[27,28]. The sulfonate groups are neutralized by another cation, often a proton, leading to the formation of sulfonic acid groups (–SO_3H)[28–30]. By spray coating a PFSA ionomer solution on the cathode (sputtered Cu on a porous PTFE gas diffusion electrode (GDE)), the –SO_3H group is expected to be preferentially exposed to the hydrophilic catalyst surface. We reasoned that exchanging the protons on the PFSA ionomers with alkali cations (thereby creating the ICM) would tailor the local microenvironment and provide a sufficiently alkali cation-rich environment at the OHP (Fig. 2c). ICM alters local pH by excluding anions and accumulating CO_2R-generated OH$^-$ at the catalyst surface. The negatively charged sulfonate groups in ICM inhibit OH$^-$ transport

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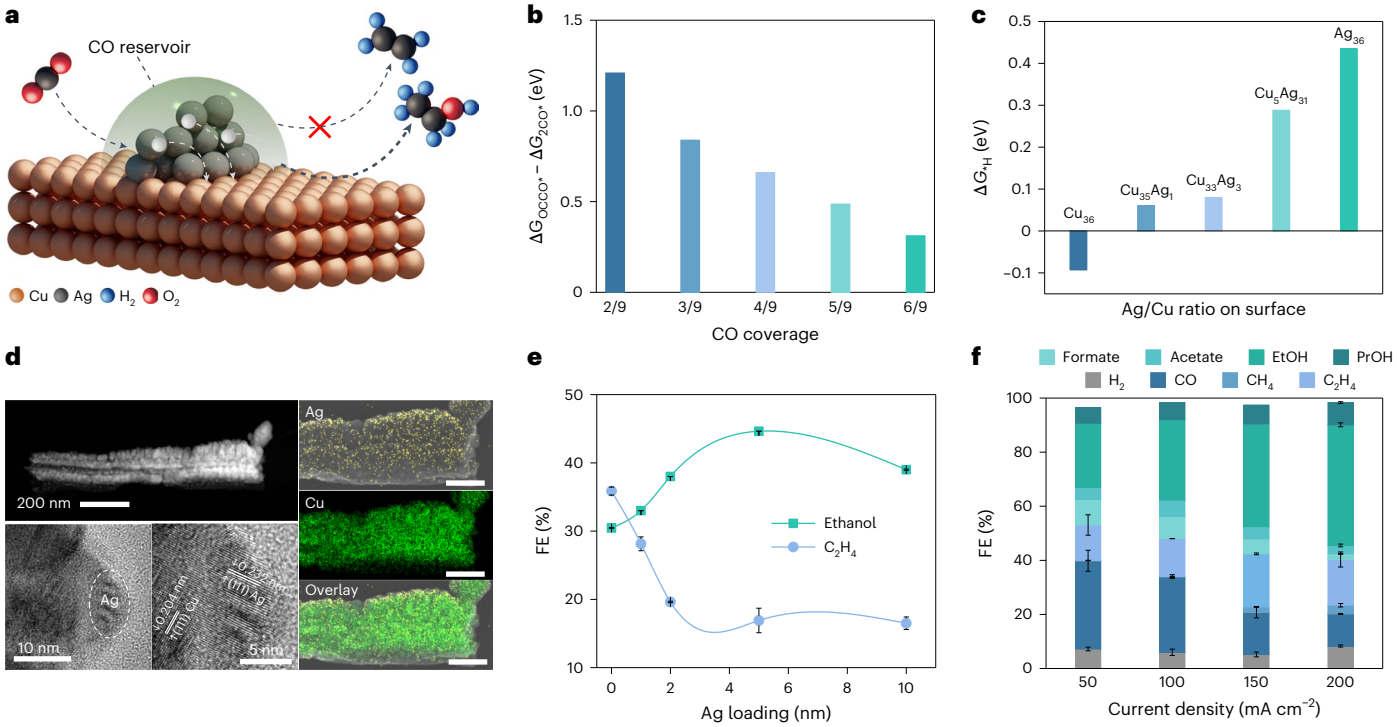

Fig. 3 | Design of a Cu–Ag catalyst for enhanced ethanol production. **a**, A schematic of a Cu–Ag catalyst where Ag clusters (grey) on the Cu surface (orange) can aid in increasing the local concentration of CO, while weakening H* binding (white) and lowering HER activity. **b**, The C–C coupling free energy of Cu under different CO* coverages. **c**, The hydrogen binding energy on Cu, Ag and CuAg alloys at different Ag contents. **d**, STEM with the corresponding EDS elemental mapping of Ag and Cu as well as a HRTEM image of Cu–Ag on a PTFE nanofibre. The loading of Ag is 5 nm. The scale bar in EDS images represents 100 nm. **e**, Ethanol and ethylene FE at different loadings (measured through sputter deposition thickness) of Ag on top of a Cu GDE. The data were obtained at a current density of 200 mA cm^{-2} and an anolyte composition of 0.01 M H_2SO_4. **f**, The FE of CO_2R products (carbon monoxide, methane, formate, ethylene, ethanol (EtOH), *n*-propanol (PrOH) and acetate) and H_2 at different current densities in a Cu–Ag-based MEA equipped with ICM. The loading of Ag is 5 nm. For **e** and **f**, the plotted values are means of three replicates, while error bars indicate the standard deviation.

through the Donnan exclusion effect. This boosts local OH^- concentration, increasing the local pH at the catalyst surface and favouring CO_2R (Fig. 2c)[31,32]. High-resolution transmission electron microscopy (HRTEM) revealed the presence of a 5–7 nm homogeneous and conformal coating of ICM over the catalyst (Fig. 2d). Scanning transmission electron microscopy (STEM) with the elemental energy-dispersive X-ray spectroscopy (EDS) mapping confirmed a homogeneous distribution of potassium over the entire catalyst, establishing the ICM (Supplementary Fig. 7). Cations are expected to accumulate at the OHP with decreasing concentration through the ICM, which results in a cation accumulation distance that is much wider than the subnanometre distance estimated without an ICM (Fig. 2b).

Multi-carbon production at high energy and carbon efficiency

To probe the efficacy of the ICM in promoting C–C coupling, we studied CO_2R in an acid-fed MEA electrolyser. The Cu catalyst on GDE was equipped with an ICM and AEL. The AEL is sprayed on top of the ICM at a thickness of about 5 µm (Supplementary Fig. 8). We first optimized the cation availability on the catalyst surface by tuning the cation loading in the ICM (for example, at a cation:PFSA blend ratio of 2). Increasing the cation loading from 0 to 4 µl cm^{-2} decreased the H_2 FE from 40% to 10% while improving the ethylene FE from 15% to a peak of 35% (Fig. 2e and Supplementary Fig. 9). Previous studies attribute the enhanced activation of CO_2 to electrostatic interactions with adsorbate dipoles or changes in surface charge density[22,33,34]. High concentrations of available cations in the electrical double layer (enabled by the ICM) and high pH (enabled by the AEL) are expected to stabilize CO_2 intermediates on the catalyst surface (Supplementary Figs. 10 and 11) and decrease the thermodynamic energy barrier to multi-carbon production[22,33,34]. In contrast to H^+ ions, cations result in less osmotic drag and less water at the catalyst surface[21]. Assuming water to be the proton source for HER[33,35] and that HER is unaffected by cation identity[34,36], a lower amount of water present, owing to the hydrophobic nature of the ICM, could be another reason for the low H_2 FE observed. Reduced water transport could also contribute to less flooding in the catalyst layer pores, which decreases the thin film thickness through which CO_2 diffuses from the gas phase to the catalyst surface and thereby accelerates CO_2R[37–39]. ICM loadings higher than 4 µl cm^{-2} decreased ethylene formation and promoted HER. At increasing ICM loading, a higher cation concentration is expected to be distributed on the catalyst surface which can reduce CO_2 solubility (via the Sechenov effect)[40,41]. This results in a higher amount of solvated water, which can also decrease the CO_2 concentration on the cathode surface[40,41]. We also examined the effect of cation identity in the ICM. At the same applied current density (200 mA cm^{-2}) and cation concentration (4 µl cm^{-2}), we noticed that Na^+ caused the highest H_2 FE and the lowest ethylene FE (Supplementary Fig. 12), K^+ showed the opposite performance, and Cs^+ exhibited a similar performance to that of K^+. This behaviour can be explained by the solvation size and the hydration number of these cations increasing in the following order: $Cs^+ < K^+ < Na^+$ (ref. 21). Cs^+ and K^+ are expected to reach a higher concentration at the OHP (Fig. 2b). Consequently, Cs^+ and K^+ result in the highest electric field strength (Fig. 2a)[21] and CO_2R rate[22]. The ICM containing K^+ was employed for subsequent testing. The product distribution for the Cu catalyst with the ICM revealed that C_{2+} generation can reach 75% FE at 200 mA cm^{-2} (Fig. 2f).

We assessed the ICM-equipped Cu catalyst carbon efficiency performance by constraining CO_2 availability (Supplementary Fig. 13). By reducing the CO_2 flow rate from 5.0 to 0.8 standard cubic centimetres per minute (SCCM) while operating at 200 mA cm^{-2}, the carbon

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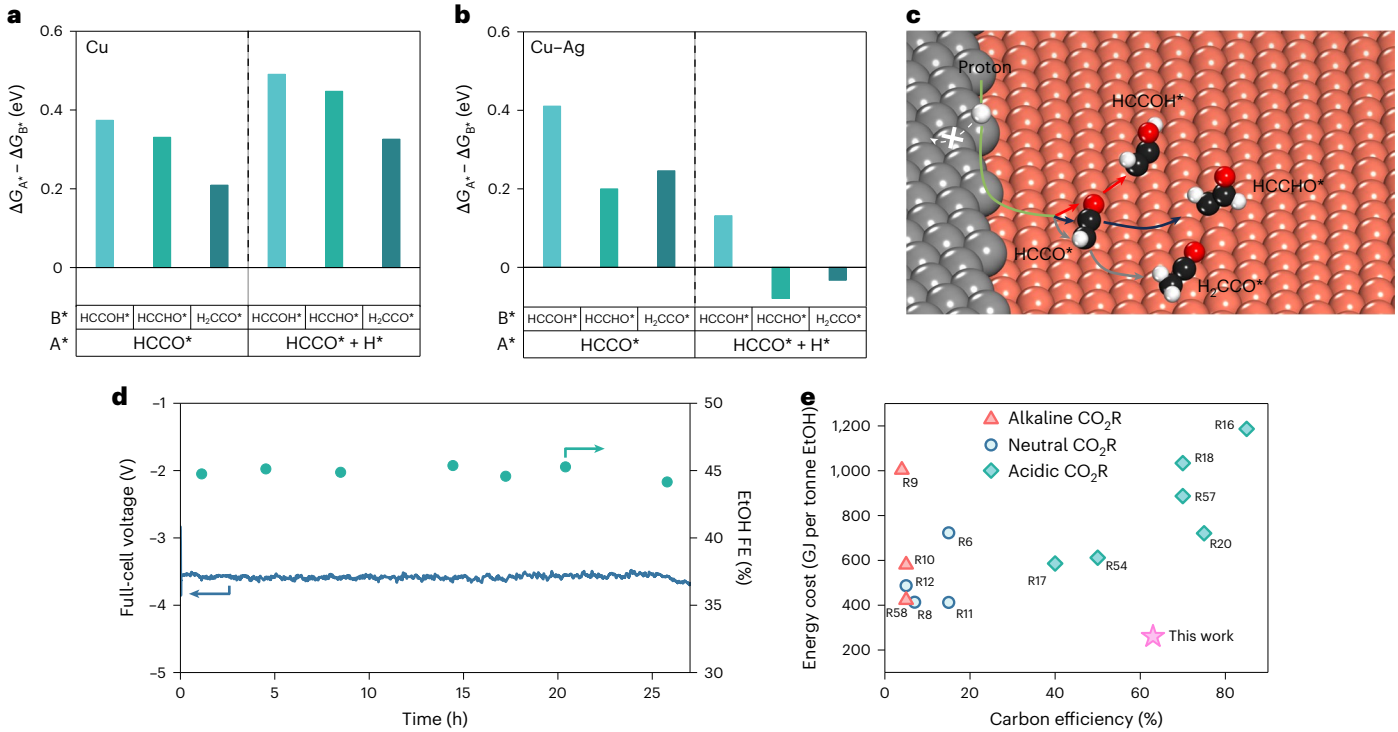

Fig. 4 | Mechanism of selective ethanol production and stable performance of ICM-equipped Cu–Ag catalyst in an MEA. a, The Gibbs free energy of HCCO* protonation on Cu via a PCET (left) or H* spillover mechanism (right). **b**, The Gibbs free energy of HCCO* protonation on Cu–Ag via a PCET (left) or H* spillover mechanism (right). **c**, Schematic illustrating the spillover mechanism of H* in a Cu–Ag catalyst, at the selectivity-determining step. Colour representation: Ag, grey; Cu, orange; C, black; O, red; H, white. **d**, CO_2R stability experiment for ethanol (EtOH) production with a Cu–Ag catalyst at 5 nm Ag loading and 200 mA cm^{-2}. The blue arrow corresponds to full-cell voltage and the teal arrow corresponds to the ethanol FE. **e**, A comparison of carbon efficiency and the ethanol production energy cost between the ICM-equipped Cu–Ag catalyst and the state-of-the-art CO_2R electrolysers used for ethanol production. The comparison is limited to reports implementing an MEA with a total current density of ≥100 mA cm^{-2} (refs. 6,8–12,16–18,20,54,57,58). R, reference.

efficiency increased from 11% to 61%. To enhance industrial viability of CO_2 electrolysers, reducing energy consumption is crucial. We then focused on improving energy efficiency at an industrially relevant current density (200 mA cm^{-2}). Using electrochemical impedance spectroscopy, we identified high ohmic losses, primarily from membrane and contact resistance. Thinner CEM membranes substantially reduced full-cell voltage, especially at higher current densities (Supplementary Note 3). By employing a catalyst-coated membrane, we achieved substantial voltage savings, crucial for commercial CO_2R feasibility. This approach was adopted for subsequent experiments. (Supplementary Note 3). The achieved $EE_{C_{2+}}$ (25%) represents a 2-fold increase compared with the best previous reports of MEA CO_2 electrolysers operating in a similarly high carbon efficiency regime (>60%) (Fig. 2g)[16,17,42]. The observed enhancement of C_{2+} production at high carbon efficiency in the case of the ICM-equipped Cu catalyst can be explained by sustained CO_2 availability (due to the hydrophobic nature of PFSA) and increased C–C coupling resulting from the local enrichment of the catalyst surface with alkali cations.

Bifurcation of the ethylene and ethanol pathways

To further enhance the selectivity of C_{2+} products, we sought to incorporate HER-suppressing Ag islands atop the Cu. We hypothesized that Ag clusters can act as CO-producing centres on the Cu surface while maintaining a low H* coverage, therefore limiting HER. A higher local CO production can increase the CO* coverage on Cu and facilitate the production of key C_2 intermediates (Fig. 3a)[43–47]. Using density functional theory (DFT) calculations, we found a decrease in the dimerization energy for C–C coupling when a higher concentration of CO exists on the Cu surface (Fig. 3b). To understand the influence of Ag on the coverage of H* on Cu surface, we theoretically investigated the hydrogen binding energy on Cu, Ag and Cu–Ag surface alloys of different composition and found that, upon incorporating Ag on Cu slabs, the binding energy and, by extension, the surface coverage of H* decrease substantially (Fig. 3c). The coverage of H* is directly related to HER activity through the rate-determining Volmer step, and its diminishing coverage can signal HER suppression. These findings suggested that decorating Ag on Cu surfaces tends to promote CO production and lower HER activity.Therefore, we experimentally designed a family of Cu–Ag catalysts in which pure Cu with a corrugated surface was sputtered with different loadings of Ag, yielding a surface with mixed Cu and Ag sites (Fig. 3d). STEM revealed the homogeneous distribution of Ag over the Cu surface. HRTEM images of Cu–Ag on PTFE further show uniformly decorated nanoclusters of Ag on the Cu surface, establishing extensive Cu–Ag interfaces (further characterizations are shown in Supplementary Figs. 14–21).

To assess the impact of the Ag loading, we evaluated the electrochemical performance of the ICM-equipped Cu–Ag catalyst at different Ag loadings in the acid-fed MEA electrolyser (Fig. 3e,f). Increasing the Ag loading on the Cu surface boosted ethanol production, with a concurrent decrease in ethylene formation. A Cu catalyst with 5-nm-thick Ag reached a peak ethanol FE of 45%, the highest achieved ethanol FE in acid-fed CO_2 electrolysers at an industrially relevant current density (200 mA cm^{-2}; Supplementary Table 1). We find that the previously well-understood pH independence of ethylene and ethanol formation no longer holds (Supplementary Fig. 22)[48–51]. This prompts the presence of a proton-transfer elementary step before the rate-determining step. We propose that the H* present at low coverages on Ag can transfer to the selectivity-determining C_2 intermediate on Cu at the Cu–Ag interface. As the H* formation on Ag is pH dependent, the change in ethylene versus ethanol selectivity becomes pH dependent. To probe

this hypothesis, we compare the H* spillover-assisted pathway on Cu and Cu–Ag at the selectivity-determining step.

It is accepted that CCO* is a key intermediate along the C_2 product pathways, and the proton-coupled electron transfer (PCET) on HCCO* can determine whether the ethanol or ethylene pathway will be followed[52]. Hence, we compare the hydrogenation of the oxygen, centre carbon and Cu-anchored carbon in HCCO* via direct PCET and a H* spillover mechanism on both Cu and Cu–Ag slabs (Fig. 4a,b and Supplementary Note 4)[43,53]. The protonation of the oxygen in HCCO* can lead to intermediates lying along both ethylene and ethanol pathways (HCCOH*), while the protonation of either the centre carbon resulting in HCHCO* or the Cu-anchored carbon resulting in H_2CCO* produces intermediates that lie exclusively along the ethanol pathway[52]. On Cu, the PCET pathway is more energetically favoured than the H* spillover pathway (Fig. 4a). The H* coverages on Cu are expected to be higher than those on Ag owing to the higher binding energy, and hence, the H* will facilitate the lower barrier HER reaction and not the energetically less favoured H* spillover. On the Cu–Ag system, the H* spillover pathway is more energetically accessible than the PCET pathway (Fig. 4b). Moreover, the spillover pathway renders the CHCHO* and H_2CCO* formation pathways exergonic, while the ethylene-producing HCCOH* remains endergonic. The result of this would be a shift in the selectivity-determining step towards ethanol formation, through a H* spillover mechanism (Fig. 4c). The Cu–Ag system is modelled with an Ag monolayer on a Cu surface with sufficient Cu empty sites for reaction intermediates, and the system closely mimics the Ag clusters present on the Cu–Ag catalyst. Higher loadings of Ag result in more Cu–$Ag_{cluster}$ boundaries, areas that were found to enable the H* spillover mechanism. The calculations also demonstrated that C–O bond scission is thermodynamically unfavourable on the interface of Cu–Ag clusters, which would favour ethanol production over ethylene (Supplementary Fig. 23). These findings further support the trend of ethylene and ethanol formation on Cu–Ag catalysts as a function of Ag loading (Fig. 3e), where the higher Ag loading resulted in more Cu–$Ag_{cluster}$ boundaries and higher ethanol FE, in parallel with ethylene suppression (Supplementary Fig. 24).

We also assessed the operating stability of the Cu–Ag catalyst equipped with an ICM in a f-BPM acidic MEA electrolyser (Fig. 4d). The electrolyser retained its initial ethanol FE (45%) after 26 h of continuous operation (Supplementary Fig. 25). By progressively reducing the flow rate of CO_2 from 5 to 0.8 SCCM, we achieved a carbon efficiency of about 63% towards CO_2R products (Supplementary Fig. 26). A carbonate-free electrolyser combined with the simultaneous achievement of high ethanol FE and high carbon efficiency resulted in a total production energy cost of 260 GJ per tonne ethanol. The ethanol production energy cost and carbon efficiency obtained through the ICM Cu–Ag approach outperform those of previous CO_2-to-ethanol reports (Fig. 4e and Supplementary Tables 1 and 2)[6,10–12,45,54]. As we move towards scaling up CO_2 electrolysers, we will indeed need to address challenges such as pressure differences, heterogeneity and heat management.

Conclusion

In this work, we present a catalyst microenvironment design principle for an acid-fed MEA electrolyser. Our findings demonstrate that enriching the catalyst surface with alkali cations while trapping locally generated OH^- using the ICM strategy enables the simultaneous achievement of high energy efficiency (25% towards C_{2+} products) and carbon efficiency (61%) in an acid-fed MEA electrolyser. By designing Cu–Ag catalysts and implementing the ICM strategy, we achieved electrosynthesis of ethanol in an MEA electrolyser with an FE of 45%, carbon efficiency of 63% and energy cost of 260 GJ per tonne ethanol simultaneously. This work presents CO_2 conversion to ethanol, with performance approaching industrial standards.

Methods

Chemicals and materials

Sulfuric acid (H_2SO_4; 98%), potassium sulfate (K_2SO_4; reagentPlus, ≥99.0%), potassium hydroxide (KOH; ACS Reagent, ≥98%, pellets), sodium hydroxide (NaOH; ACS Reagent, ≥85%, pellets), caesium hydroxide (CsOH; ACS Reagent, ≥98%, pellets), methanol, acetone and ethanol were purchased from Sigma-Aldrich. The anion-exchange ionomer powder (Aemion, AP2-CNN5-00-X) was purchased from Ionomr Innovations. All the Nafion membranes (117, 212 and XL) and PFSA dispersion (PFSA Dispersion (D5, 5%), alcohol based) were purchased from Fuel Cell Store. Titanium grade 2 felt, which was 60% porous before coating, was used to make the anode. The anode was coated with IrO_2 (Evoque, Magneto Special Anodes). Silver nanoparticles (Ag, 99.99%, 20 nm, metal basis) were acquired from US Research Nanomaterials. For the preparation of all electrolytes, deionized water (18.2 MΩ) was used. To sputter Cu catalyst onto the PTFE substrate (pore size of 450 nm, with a polypropylene support on the backside), a Cu target (>99.99%) was used in an Angstrom Nexdep sputtering system with a constant sputtering rate of 1.0 Å s^{-1} under vacuum (~10^{-6} Torr) until a catalyst thickness of 200 nm was achieved. The Cu–Ag catalysts were prepared by sputtering different Ag thicknesses on top of the Cu surface using an Ag target (>99.99%) with a constant sputtering rate of 0.1 Å s^{-1}. The nominal sputtering thickness of Ag was considered as Ag loading throughout the work.

GDE preparation

The GDEs (cathodes) equipped with an AEL were prepared by spraying a dilute Ameion solution to achieve the desired loading (3 mg cm^{-2}). The Ameion solution was prepared by dissolving 158 mg of the ionomer powder in 40 ml of a mixture of methanol and acetone (80 vol% + 20 vol%, respectively). This solution was diluted by doubling the total volume of the dispersion (with the same amount of methanol and acetone) before spraying and dried overnight before usage. To prepare cathodes equipped with ICM and the AEL, a desired amount of activated PFSA dispersion was spray coated on the cathode surface. The PFSA dispersion was prepared by diluting 40 µl of PFSA solution (5 wt%) in 2 ml methanol. To activate the PFSA with the cation, 80 µl of 1 M KOH, 1 M CsOH or 1 M NaOH was added to the PFSA dispersion and mixed for 10 min. This dispersion was stored overnight before the spray. After spray coating the ICM layer, the sample was allowed to air dry for at least 4 h before the AEL was sprayed.

The PFSA dispersion (5 wt% in solution) used in this work has the following properties: chemical formula $(C_7HF_{13}O_5S{\cdot}C_2F_4)_x$; molecular weight 554 g mol^{-1} and density 0.93 g ml^{-1}. The mass of PFSA on the cathode surface, given a PFSA loading of 2 µl cm^{-2}, can be estimated as

$$m_{\mathrm{PFSA}} = 0.002\,\frac{\mathrm{ml}}{\mathrm{cm}^2} \times 0.93\,\frac{\mathrm{g_s}}{\mathrm{ml}} \times \frac{5}{100}\,\frac{\mathrm{g_{PFSA}}}{\mathrm{g_s}} = 0.093 \times 10^{-3}\ \mathrm{mg\ cm^{-2}}.$$

The mass fraction of sulfonic acid groups (x_s) within the PFSA can be calculated as follows, considering 1 mol of PFSA as the reference and a molecular weight of 81 g mol^{-1} for a sulfonic group ($-SO_3H$):

$$x_{\mathrm{s}} = \frac{81\ \mathrm{g\,mol^{-1}}}{554\ \mathrm{g\,mol^{-1}}} = 0.15.$$

The number of moles of sulfonic acid groups is determined as

$$n_{\mathrm{s}} = \frac{9.3 \times 10^{-5}\ \mathrm{g\ cm^{-2}} \times 0.15}{81\ \mathrm{g\,mol^{-1}}} = 1.72 \times 10^{-7}\ \mathrm{mol\ cm^{-2}}.$$

This represents the lower limit of K^+ that is used to neutralize the sulfonic acid groups (1.72×10^{-7} M K^+ on the cathode surface). Considering 2 µl of PFSA solution, 4 µl of 1 M KOH was added to neutralize the sulfonic acid groups. Assuming all K^+ was spray coated on the Cu

surface, 4.0×10^{-6} mol cm^{-2} of K^+ was sprayed onto the catalyst surface, representing an excess amount used to stoichiometrically neutralize 1.72×10^{-7} mol cm^{-2} of sulfonic acid groups. Thus, the upper limit of K^+ concentration on the cathode surface can be considered as 4.0 mM.

Materials characterization

Scanning electron microscopy imaging and EDS X-ray mapping were performed via an Ultra-High-Resolution Schottky SEM SU7000, Hitachi microscope. A field emission transmission electron microscope (HF3300, Environmental-CFE-TEM) was used to capture the STEM, HRTEM images and EDS elemental mapping of Cu electrodes equipped with ICM and coated with Ag clusters. The crystalline structure of catalysts was examined with Rigaku Miniflex 600 6G Benchtop powder X-ray diffraction (step 0.01°, speed 1.5° min^{-1}). A Renishaw inVia Raman microscope equipped with a water immersion objective was employed for operando Raman spectroscopy. The spectra were obtained by employing a 785 nm laser, with a laser intensity set to 0.05% full laser intensity. The integration time was 0.1 s, and the measurements were repeated 100 times. For the measurements, an open-structured flow cell was utilized. As for the electrodes, an Ag/AgCl electrode filled with a saturated aqueous KCl solution served as the reference electrode, while a platinum wire was used as the counter electrode.

The in situ and ex situ X-ray absorption spectroscopy (XAS) measurements were conducted at the Hard X-ray Micro-Analysis Beamline at the Canadian Light Source. The in situ XAS spectra of the Cu K edge were recorded in fluorescence mode using a custom in situ XAS flow cell. The ex situ spectra were recorded in transmission mode. The energy calibration was performed using standard metallic Cu foil (8,979 eV). During measurements, 0.01 M H_2SO_4 + 0.1 M KCl aqueous solution was applied for the catholyte and 0.01 M H_2SO_4 for the anolyte. A Nafion 117 CEM separated the two electrolyte compartments. The measured catalysts included sputtered Cu (200 nm) with different loadings of silver on a PTFE substrate. A Pt anode and an Ag/AgCl (saturated KCl) electrode were used as the counter and reference electrodes, respectively. The electrolytes were flowed using peristaltic pumps at a flow rate of 1 ml min^{-1}. Pure CO_2 gas was supplied to the cathode during test.

MEA and CO_2 electrolysis

All CO_2R experiments were performed in a 1 cm^2 cell prepared with a grade 2 titanium cathode and anode flow fields. To construct the f-BPM, a commercial Nafion membrane was used as the CEM on top of the prepared AEL. IrO_2 felt was used as the anode. Electrochemical measurements were performed with an Autolab electrochemical workstation (PGSTAT204 equipped with a 10A booster). Throughout all experiments, unless otherwise specified, CO_2 was fed to the electrolyser at 5 SCCM using a mass flow controller. The anode side of the electrolyser was fed with 0.01 M H_2SO_4 for all experiments, unless otherwise specified. The cell voltages reported in this work are full-cell voltages without *iR* correction (*i*, current; *R*, resistance). For the flow-cell setup, the Cu–Ag catalyst GDE was used as the working electrode (cathode), an IrO_2 felt as the counter electrode (anode) and an Ag/AgCl electrode as the reference electrode. A CEM (Nafion 117) was used to separate the anodic and cathodic compartments. The catholyte and anolyte were 0.01 M H_2SO_4 + 3 M KCl and 0.01 M H_2SO_4, respectively. The stability experiment involved gradually increasing the applied current, starting at 100 mA cm^{-2} for an initial 10 min, followed by a transition to 200 mA cm^{-2} for the remaining duration.

CO_2R product analysis

The gas outlet stream (containing CO_2R products and unreacted CO_2) was sampled for product analysis. A gas chromatograph (PerkinElmer Clarus 590) coupled with a thermal conductivity detector and flame ionization detector was utilized for this purpose. The chromatograph contains a Molecular Sieve 5A Capillary Column and a packed Carboxen-1000 Column, with argon serving as the carrier gas. The flow rates of gas in and out of the cell were determined via a bubble column. The FE of gas products was determined using

$$\mathrm{FE_{gas}}\,(\%) = \frac{z \times r \times F \times v}{I \times V_\mathrm{m}},$$

where z corresponds to the number of electrons involved in the production of the gas product in question, r represents the concentration of detected gas in ppm, F is the Faraday's constant (96,485 C mol^{-1}), v is the gas flow rate, I represents the total current and V_m is the unit molar volume of gas.

Proton nuclear magnetic resonance spectroscopy was utilized to measure the quantities of liquid products. This was performed on an Agilent DD2 500 spectrometer while in D_2O water suppression mode. Dimethyl sulfoxide was employed as an internal standard. The FE of liquid products was determined using

$$\mathrm{FE_{liquid}}\,(\%) = \frac{z \times F \times n_\mathrm{product}}{I \times t},$$

where n_product is the total quantity of products measured in moles, and $I \times t$ (C) denotes the total charge (Q) that was passed during the experiment.

Carbon and energy efficiency calculation

Carbon efficiency (or SPCE; the percentage of CO_2 converted per total CO_2 input), which pertains to gas, liquid or a combination of gas and liquid products at 25 °C and 1 atm, was determined via

$$\text{Carbon efficiency}\,(\%) = \frac{(j_\mathrm{product} \times 60\ \mathrm{s})/(n \times F)}{(v \times 1\ \mathrm{min})/V_\mathrm{m}},$$

where j_product is the partial current density of a specific group of CO_2R products (measured in amperes), v is the gas flow rate (l min^{-1}) and V_m is equivalent to 24.05 (l mol^{-1}). The full-cell EE for each product was determined using

$$\mathrm{EE_{product}} = \frac{\left(1.23\ \mathrm{V} + \left(-E^0_\mathrm{product}\right)\right) \times \mathrm{FE_{product}}}{-E_\mathrm{cell}}.$$

Here, E^0_product refers to the thermodynamic potential for the formation of the specific CO_2R product, and E_cell is the full-cell voltage obtained from the MEA electrolyser without *iR* correction.

Multiphysics modelling

The local concentrations of cations at different current densities, the local pH at the cathode surface with different configurations, and the electric field at the electrode surface were modelled in COMSOL (COMSOL Multiphysics version 5.5). A simulation tutorial was provided in our previous report[55]. See Supplementary Information for more details.

DFT calculation

DFT was applied to calculate the Gibbs free energy diagrams of CO_2R and HER on the (111) surfaces of Cu (Cu(111)) and Ag clusters on Cu surface (Ag–Cu(111)) as well as the charge transfer between the HCCOH* intermediate and surface. All the DFT calculations were carried out in the Vienna ab initio simulation package with a plane wave pseudo-potential implementation. See Supplementary Information for more details.

Data availability

All data supporting the findings of this study, including computational specifics, and other experimental and microscopic analyses, can be found in the Article and its Supplementary Information.

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Author contributions

D.S. and E.H.S. supervised the project. A.S.Z. conceived the idea and performed the electrochemical experiments and analyses. A.S.Z. and T.A. co-wrote the manuscript. F.L. performed the DFT calculations. F.L. and T.A. carried out the FEM simulation. R.D. assisted with the DFT calculation and manuscript writing. E.S., F.A., C.P.O., R.K.M. and D.Y. helped with electrochemical experiments. J.K. assisted with TEM imaging. A.O. and Y.Z. assisted with the catalyst preparation. P.P., D.K. and S.P. carried out the XAS measurements. M.Z. assisted with the schematics. L.F. carried out the X-ray diffraction measurement. C.M.G., J.P.E. and A.H.I. assisted with the research advice, discussions and editing. All authors contributed to the manuscript editing.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to David Sinton.

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